

Dataset Analysis and Insights

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sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.



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To address your request, I will first analyze the dataset to identify its key characteristics. Then, I will await a specific analytical question to provide a targeted answer and the corresponding Python code. Let's begin with the dataset analysis.

Step 1: Dataset Analysis

The dataset provided is named `sicor_municipal_contracts.csv`. To understand its structure and contents, I will:

1. Load the dataset using Python's `pandas` library.
2. Examine its shape (number of rows and columns).
3. Inspect the column names and data types.
4. Check for missing values.
5. Summarize key statistics for numerical and categorical columns.

Here is the Python code to perform this analysis:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Shape:", data.shape)
9 print("\nColumn Names and Data Types:")
10 print(data.dtypes)
11
12 print("\nSummary Statistics for Numerical Columns:")
13 print(data.describe())
14
15 print("\nMissing Values per Column:")
16 print(data.isnull().sum())
17
18 print("\nFirst 5 Rows of the Dataset:")
19 print(data.head())
```

Key Characteristics of the Dataset

After running the above code, I would summarize the dataset's key characteristics as follows:

1. **Shape** : The dataset contains `x` rows and `y` columns.
2. **Columns** : The dataset includes columns such as `[list column names]`, with data types ranging from numerical (e.g., integers or floats) to categorical (e.g., strings).
3. **Missing Values** : Certain columns may have missing values, which could impact analysis.
4. **Numerical Summary** : Key statistics (mean, median, min, max, etc.) for numerical columns provide insights into the range and distribution of values.
5. **Categorical Summary** : Unique values and their frequencies in categorical columns highlight patterns or categories within the data.



Step 2: Addressing a Specific Analytical Question

Please provide a specific analytical question related to the dataset. For example:

- "What is the total value of contracts awarded by each municipality?"
- "Which contractor has received the highest number of contracts?"
- "Are there any trends in contract values over time?"

Once you provide the question, I will:

1. Extract and process the relevant data using Python.
2. Provide a clear, concise answer to the question.
3. Include the Python code used to derive the answer.

Let me know your specific question so I can proceed!

How does the total amount invested vary by contract issuance month?



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To analyze how the total amount invested varies by contract issuance month, we need to:

1. Group the data by the `MonthIssued` column.
2. Sum the `InvestmentValue` column for each group.
3. Sort the results by month for clarity.

Here is the Python code to perform this analysis:

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sikor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Group by MonthIssued and sum the InvestmentValue
8 investment_by_month = data.groupby('MonthIssued')['InvestmentValue'].sum().reset_index()
9
10 # Sort by MonthIssued for better readability
11 investment_by_month = investment_by_month.sort_values(by='MonthIssued')
12
13 # Display the results
14 print("Total Investment by Month:")
15 print(investment_by_month)
```

Explanation of the Code:

1. `groupby('MonthIssued')` : Groups the data by the `MonthIssued` column.
2. `['InvestmentValue'].sum()` : Calculates the total investment value for each month.
3. `reset_index()` : Converts the grouped data back into a DataFrame.
4. `sort_values(by='MonthIssued')` : Ensures the months are displayed in ascending order.

Expected Output:

The output will be a table showing the total investment value for each month. For example:

MONTHISSUED	INVESTMENTVALUE
1	123456789.0
2	987654321.0
3	456789123.0
...	...

Answer:

After running the code, you will see the total investment values for each month. This will allow you to identify trends, such as which months have the highest or lowest investments, and understand seasonal variations in contract issuance.



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If you run the code and share the results, I can help interpret them further!

